

ENME601 Thermodynamics

Week 4 – Energy Analysis of Closed Systems

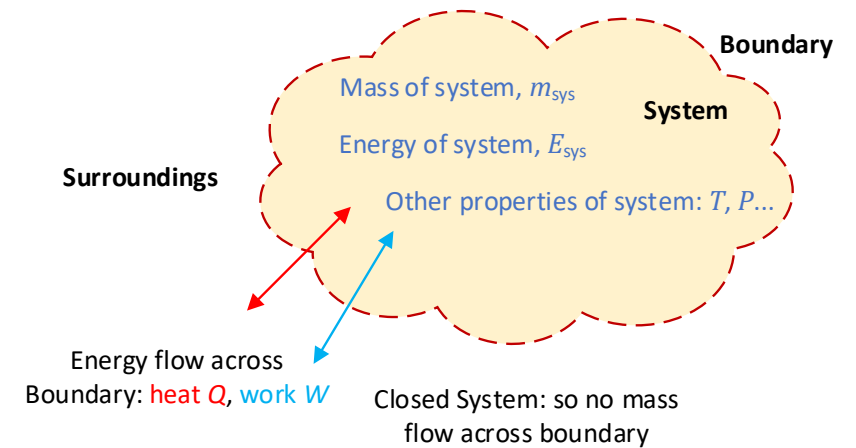
Presenter: Dr. Jay Wang

Objectives

- Examine the **moving boundary work or $P dV$ work** commonly encountered in reciprocating devices such as automotive engines and compressors.
- Develop the **general energy balance applied to closed systems**.
- Define the **specific heat at constant volume** and the **specific heat at constant pressure**.
- Relate the specific heats to the calculation of the **changes in internal energy and enthalpy of ideal gases**.
- Describe **incompressible substances** and determine the changes in their internal energy and enthalpy.
- **Solve energy balance problems for closed (fixed mass) systems** that involve heat and work interactions for general pure substances, ideal gases, and incompressible substances.

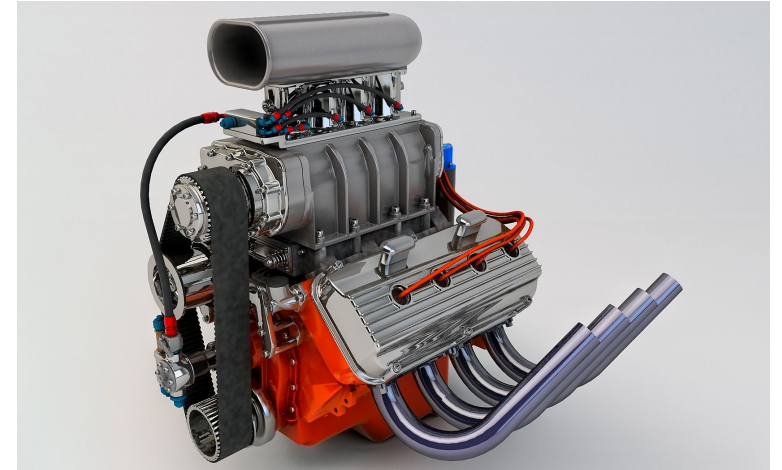
Introduction

- In Topic 2 (Chapter 2) we considered various forms of energy and energy transfer and the general form of an energy balance known as the first law.
- In Topic 3 (Chapter 3) we learned about pure substances and how to determine thermodynamic properties using tables.
- We now consider the **first law energy balance applied to closed systems** (no mass flow across boundary).
- We also consider the **specific heat capacity** and its **relationship to internal energy and enthalpy**

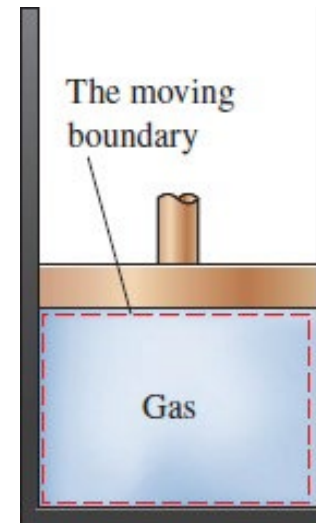


Moving Boundary Work

- One form of mechanical work often encountered in practice is associated with **the expansion or compression of a gas in a piston-cylinder device**. (e.g air compressors, internal combustion engines)
- Part of the boundary (piston face) moves back and forth - hence the term **moving boundary work** or **simply boundary work**.
- In real situations the piston and boundary are moving at high speeds (non-equilibrium and difficult to analyse).
- In this analysis we assume **quasi-equilibrium processes** which will provide **ideal values for work input and output**.



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Moving Boundary Work

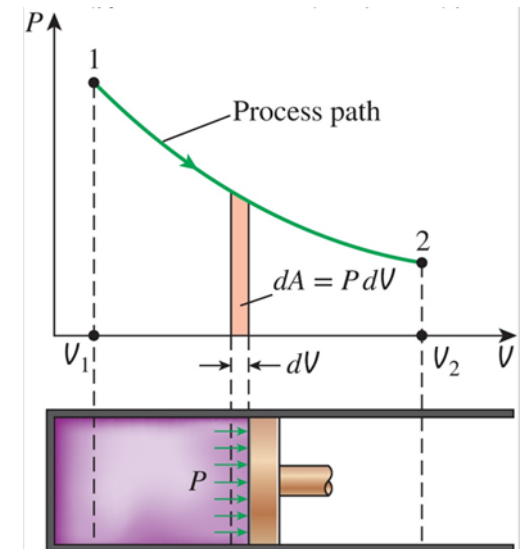
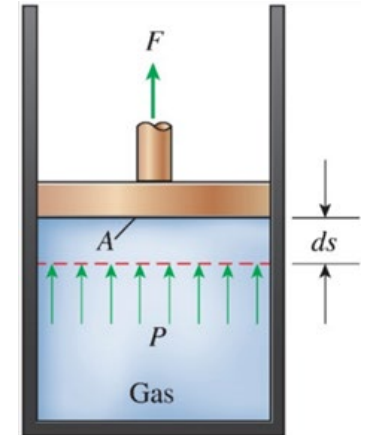
- Consider a gas enclosed in a piston cylinder device, initial gas pressure, P ; total volume, V ; and piston cross-sectional area, A .
- If the piston moves a small distance ds then the small (differential) amount of boundary work done is:

$$\delta W_b = F ds = PA ds = P dV$$

- So total boundary work over full movement of piston is:

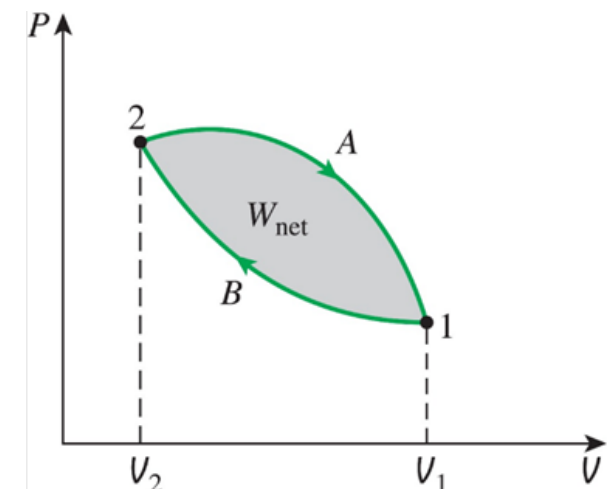
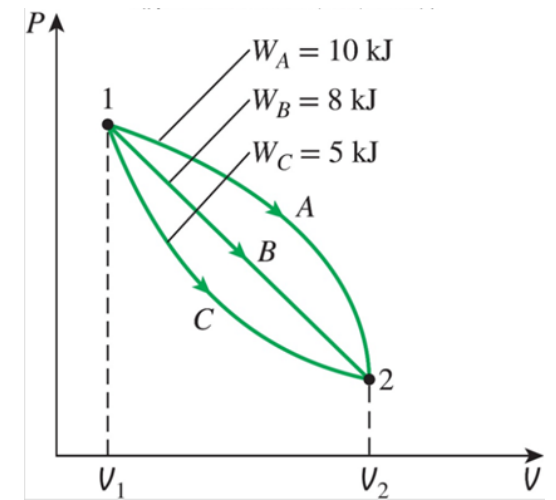
$$W_b = \int \delta W_b = \int_1^2 P dV$$

- $\int_1^2 P dV$ represents the area under the process curve (state 1 to state 2) on a $P - V$ diagram.
- So, the **ideal boundary work, W_b** is just the **area under the process path on a $P - V$ diagram.**



Moving Boundary Work

- The **ideal moving boundary work** is the **area** under the process path on a $P - V$ diagram.
- For an **expansion process** – work out $\Rightarrow W_b$ is **positive**.
- For a **compression process** – work in $\Rightarrow W_b$ is **negative**.
- For different paths between state 1 and state 2 the **boundary work will be different for each path** (since area under process path is different).
- In a cycle the *ideal* net work done, W_{net} is equal to the area enclosed by the process paths (which is equal to the difference between work done by the system and work done on the system)

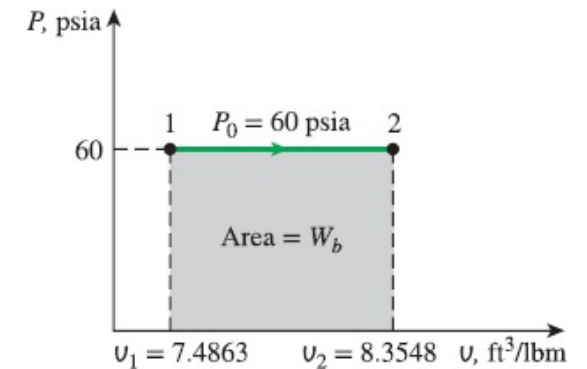


Moving Boundary Work

Constant Pressure (Isobaric) Process

$$W_b = \int_1^2 P dV \Rightarrow W_b = P(V_2 - V_1) \quad \text{Constant pressure process}$$

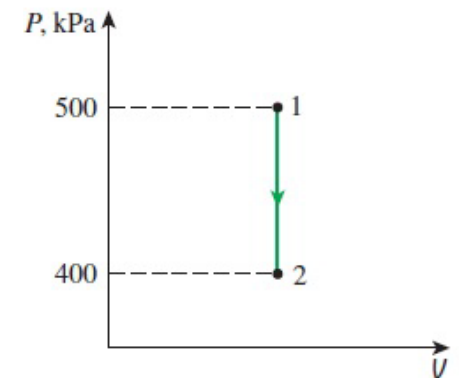
(e.g piston maintains a constant pressure on the gas or substance during the process)



Constant Volume (Isochoric) Process

$$W_b = \int_1^2 P dV \Rightarrow W_b = 0 \quad \text{Constant volume process}$$

(e.g. in case of a rigid tank, pressure changes say due to energy/heat removal but no movement of boundary so $W_b = 0$)



Moving Boundary Work

Constant Temperature (Isothermal) Process – ideal gas

- If the work of compression is all lost as heat (cooling of cylinder) or heat gain is equal to expansion work, then the temperature will remain constant during the process.
- Analysis: $PV = mRT$ and m , R and T are all constant, let $c = mRT$ so

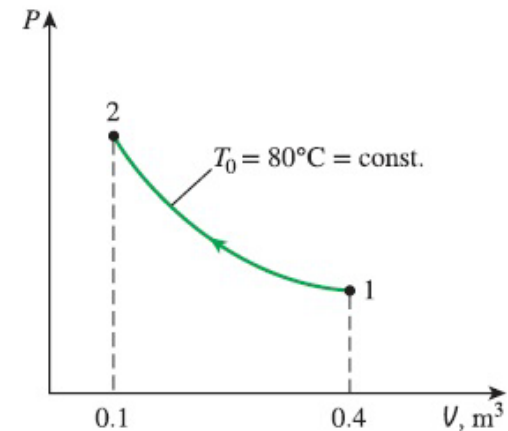
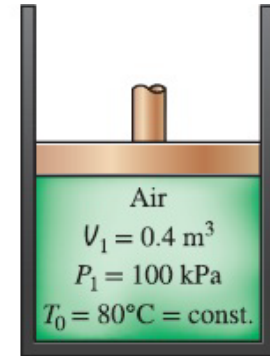
$$PV = c \quad \Rightarrow \quad P = \frac{c}{V}$$

(Or $PV^n = \text{constant}$, $n = 1$)

And **boundary work**:

$$W_b = \int_1^2 P dV = \int_1^2 \frac{c}{V} dV$$
$$\Rightarrow W_b = P_1 V_1 \ln \frac{V_2}{V_1} = P_2 V_2 \ln \frac{V_2}{V_1}$$

Constant temperature process



Moving Boundary Work

Polytropic Process – ideal gas

- During *actual* compression and expansion processes of gases there is *some* heat loss or gain in the process and it is found that the pressure and volume are related by

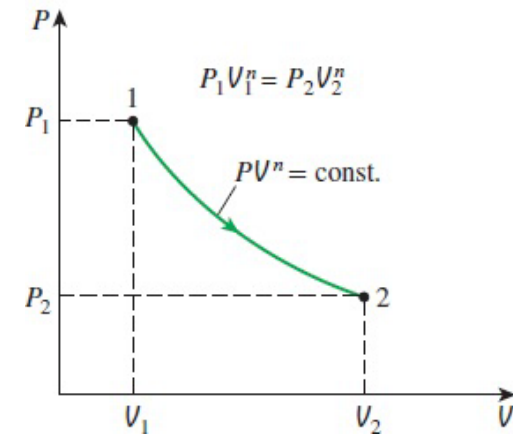
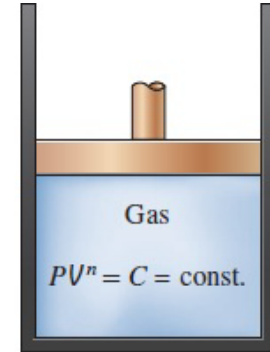
$$PV^n = \text{constant}, \quad 1 < n < k$$

- The parameter n is called the index of compression (or expansion) and k = ratio of specific heats = C_P/C_V (see later)

Analysis for **boundary work** yields

$$W_b = \frac{P_1 V_1 - P_2 V_2}{n - 1} = \frac{mR}{n - 1} (T_1 - T_2)$$

Polytropic
process



Moving Boundary Work

Adiabatic Process – ideal gas

- If the cylinder is perfectly insulated then there is no heat loss or gain during the compression or expansion process (i.e. an adiabatic process) it is found that the pressure and volume are related by

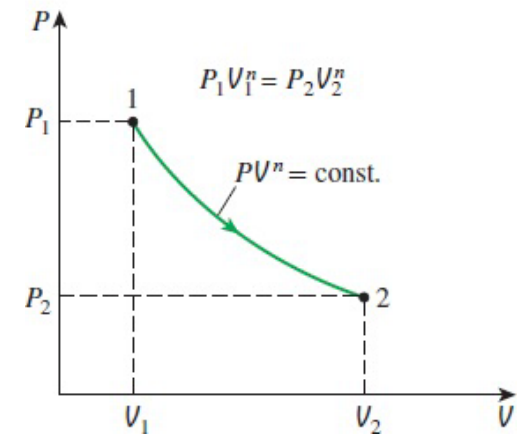
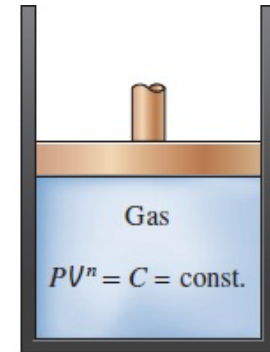
$$PV^n = \text{constant}, \quad n = k$$

- n = index of compression (or expansion) and $k = C_P/C_V$ (see later)

Analysis for **boundary work** (same as for polytropic case with $n = k$) yields

$$W_b = \frac{P_1 V_1 - P_2 V_2}{k - 1} = \frac{mR}{k - 1} (T_1 - T_2)$$

Adiabatic
process



Energy Balance for Closed Systems

- In Topic 2 (Chpt. 2) we stated the first law as a general energy balance for a system

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{net energy transfer by heat, work and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{change in internal, kinetic potential, etc energies}} \quad (\text{kJ})$$

- Or in rate form:

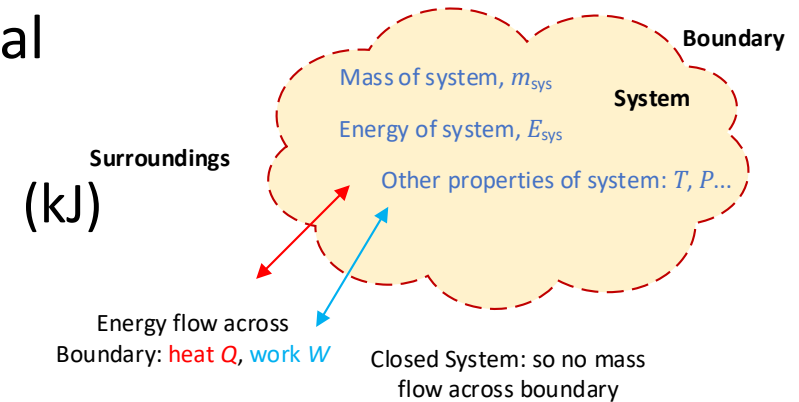
$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{rate of net energy transfer by heat, work and mass}} = \underbrace{dE_{\text{system}}/dt}_{\text{rate of change in internal kinetic, potential, etc energies}} \quad (\text{kW})$$

- For a closed system there is no energy transfer with mass so

$$(Q_{\text{in}} - Q_{\text{out}}) + (W_{\text{in}} - W_{\text{out}}) = \Delta E_{\text{system}}$$

- Or using the classical sign convention (heat in positive, work out positive) then:

$$Q - W = \Delta E_{\text{system}}$$



Energy Balance for Closed Systems

- From previous slide: $Q - W = \Delta E_{\text{system}}$
- For **stationary, simple compressible systems**:
$$(Q_{\text{in}} - Q_{\text{out}}) + (W_{\text{in}} - W_{\text{out}}) = \Delta U$$
- Or using classical sign convention (heat in positive, work out positive) for **closed systems**:

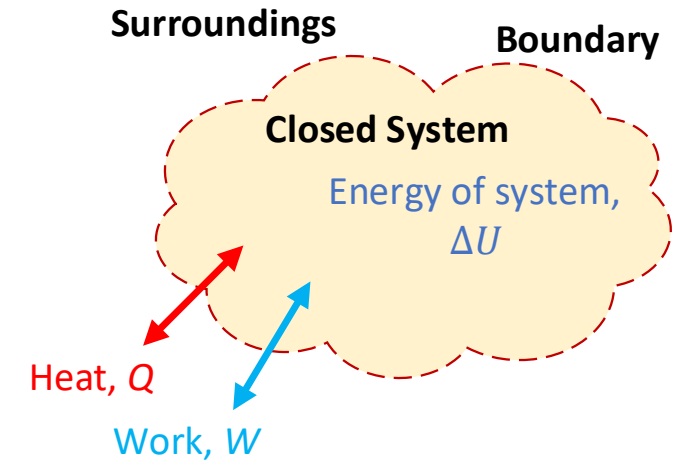
$$Q - W = \Delta U$$

Energy balance,
closed system

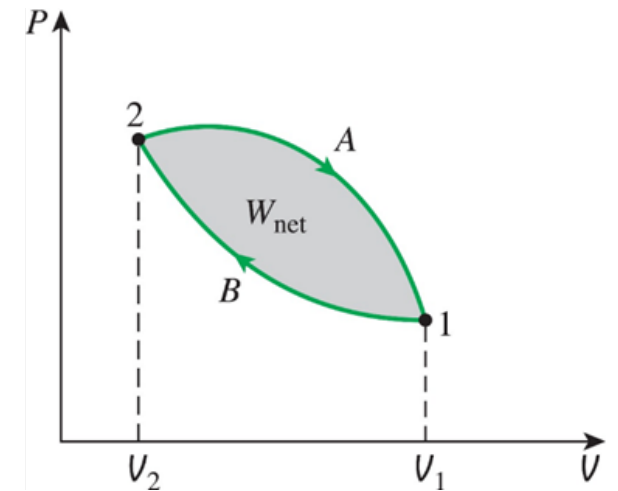
- **Cycles**

$$Q_{\text{net,in}} = W_{\text{net,out}}$$

Stationary, simple compressible system



Cycle



Energy Balance for Closed Systems

Energy balance for constant pressure expansion or compression processes.

$$Q - W = \Delta U \quad \text{First law, closed system}$$

- The work term, W is made up of boundary work, W_b and possibly other forms of work, W_{other} , so

$$Q - W_b - W_{\text{other}} = \Delta U$$

- Now for a constant pressure process, $W_b = P(V_2 - V_1)$ and $P = P_1 = P_2$, so

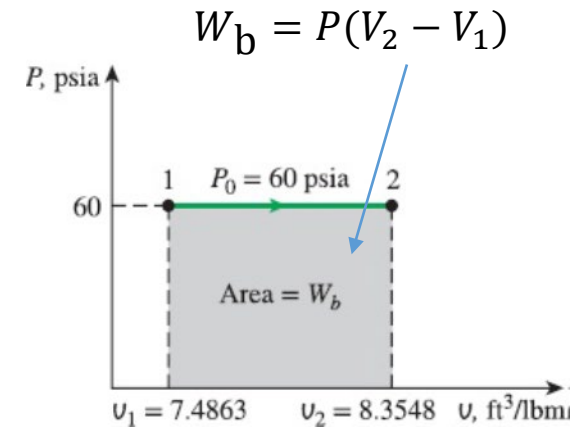
$$Q - W_{\text{other}} = \Delta U + W_b = U_2 - U_1 + P_2V_2 - P_1V_1$$

$$\Rightarrow Q - W_{\text{other}} = (U_2 + P_2V_2) - (U_1 + P_1V_1)$$

$$\Rightarrow Q - W_{\text{other}} = H_2 - H_1$$

$$\Rightarrow Q - W_{\text{other}} = \Delta H = m(h_2 - h_1)$$

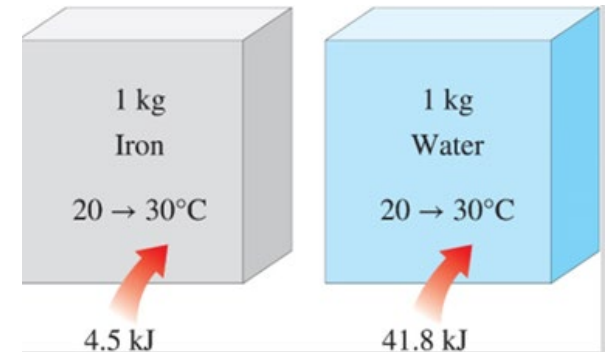
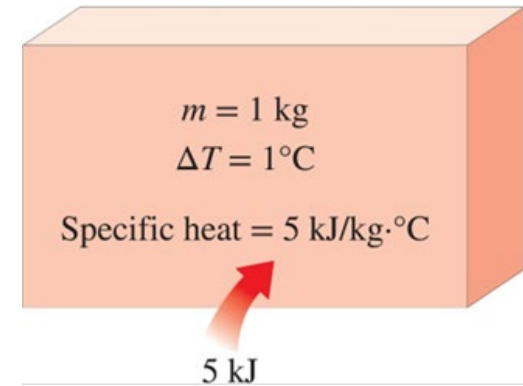
Constant pressure process
– closed system



Definition of enthalpy, H
 $H = U + PV$

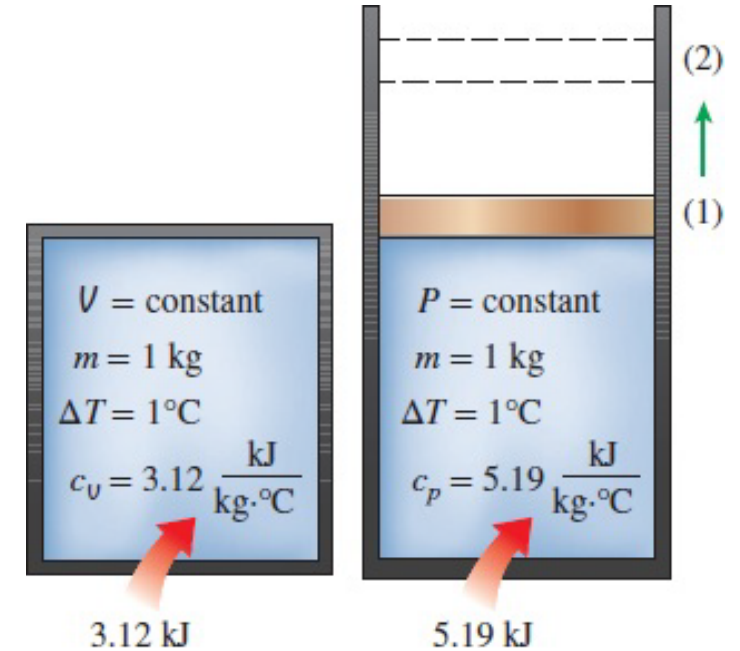
Specific Heats

- The **specific heat** is defined as the **energy required to raise the temperature of a unit mass of substance through one degree**.
- It takes different amounts of energy to raise the temperature of different substances by the same amount.
- The process of adding energy can be done at constant volume or at constant pressure:
 - Specific heat at constant volume, c_v
 - Specific heat at constant pressure, c_p
- The specific heats (c_v , c_p) are **properties** of the substance.



Specific Heats

- **Specific heat at constant volume, c_v** : the energy required to raise the temperature of a unit mass of substance through one degree as the **volume is held constant**.
- **Specific heat at constant pressure, c_p** : the energy required to raise the temperature of a unit mass of substance through one degree as the **pressure is held constant**.
- **c_p is always greater than c_v** for a substance since c_p involves additional expansion (boundary) work to maintain the substance at constant pressure.
- Units for c_v and c_p : $\left(\frac{\text{kJ}}{\text{kg}\cdot^\circ\text{C}}\right)$ or $\left(\frac{\text{kJ}}{\text{kg}\cdot\text{K}}\right)$



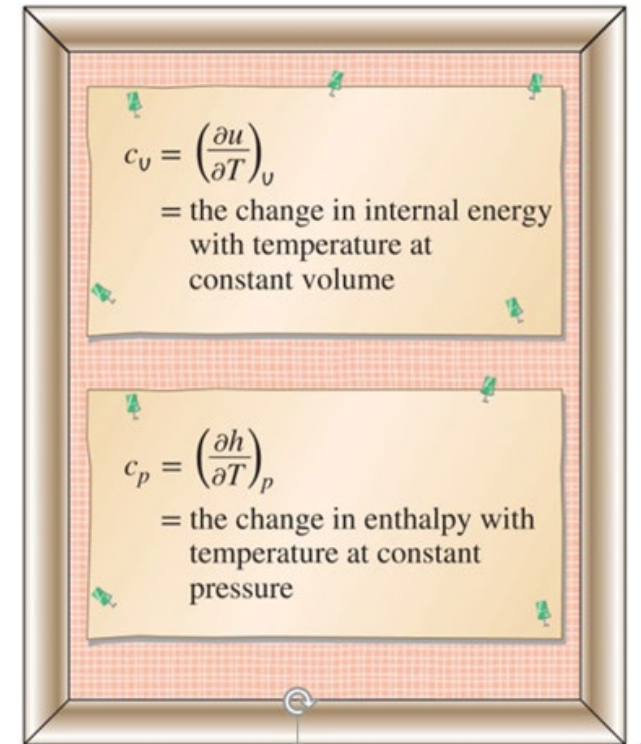
Specific Heats

- The specific heats can be **defined in terms of other thermodynamic properties**:

$$c_v dT = du \quad \text{at constant volume}$$

$$c_p dT = dh \quad \text{at constant pressure}$$

- Although **the specific heats** are defined in terms of distinct processes they **are true properties** and as such can be used in equations for any thermodynamic process. (Properties are independent of the process).
- We can use the above relations to determine the change in internal energy and enthalpy for ideal gases and for solids and liquids.



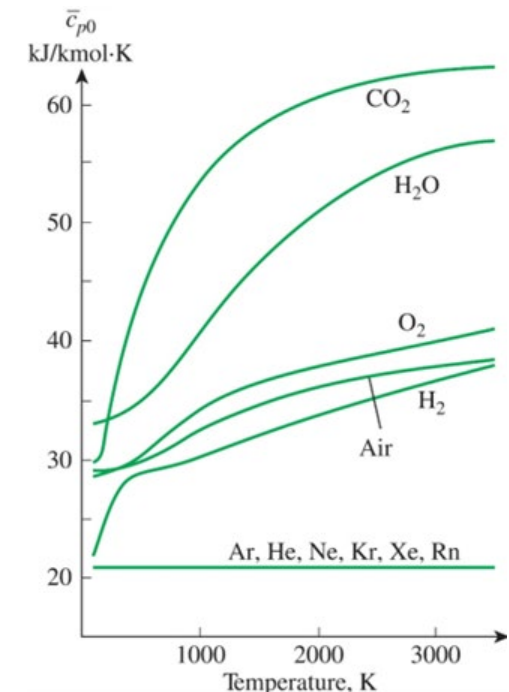
Internal Energy, Enthalpy and Specific Heats for Ideal Gases

- It can be shown that for an ideal gas the internal energy, u , the enthalpy, h and the specific heat capacities c_v and c_p **depend only on temperature**.
i.e. they are functions of temperature alone.
- u and h data has been tabulated for various gases assuming ideal gas behaviour. E.g. Table A-17 for Air. (0 K is taken as the reference temperature for these tables)

$$\begin{aligned}u &= u(T) \\h &= h(T) \\c_v &= c_v(T) \\c_p &= c_p(T)\end{aligned}$$

This is the **most accurate** way of determining u and h values for air and other gases

TABLE A-17										
Ideal-gas properties of air										
T K	h kJ/kg	P_r	u kJ/kg	v_r	s° kJ/kg·K	T K	h kJ/kg	P_r	u kJ/kg	
200	199.97	0.3363	142.56	1707.0	1.29559	580	586.04	14.38	419.55	
210	209.97	0.3987	149.69	1512.0	1.34444	590	596.52	15.31	427.15	
220	219.97	0.4690	156.82	1346.0	1.39105	600	607.02	16.28	434.78	
230	230.02	0.5477	164.00	1205.0	1.43557	610	617.53	17.30	442.42	
240	240.02	0.6355	171.13	1084.0	1.47824	620	628.07	18.36	450.09	
250	250.05	0.7329	178.28	979.0	1.51917	630	638.63	19.84	457.78	
260	260.09	0.8405	185.45	887.8	1.55848	640	649.22	20.64	465.50	
270	270.11	0.9590	192.60	808.0	1.59634	650	659.84	21.86	473.25	
280	280.13	1.0889	199.75	738.0	1.63279	660	670.47	23.13	481.01	
285	285.14	1.1584	203.33	706.1	1.65055	670	681.14	24.46	488.81	
290	290.16	1.2311	206.91	676.1	1.66802	680	691.82	25.85	496.62	
295	295.17	1.3068	210.49	647.9	1.68515	690	702.52	27.29	504.45	



Internal Energy, Enthalpy and Specific Heats for Ideal Gases

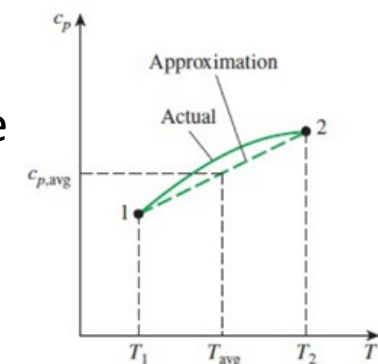
- The *change* in u or h can also be found by integrating the definition equations: $c_v dT = du$ and $c_p dT = dh$.
- If we assume that the specific heats are some constant average value over the temperature range involved, then we get:

Internal energy	Enthalpy
$\Delta u = c_{v,avg} \Delta T$	$\Delta h = c_{p,avg} \Delta T$
$u_2 - u_1 = c_{v,avg} (T_2 - T_1)$	$h_2 - h_1 = c_{p,avg} (T_2 - T_1)$
$\Delta U = mc_{v,avg} \Delta T$	$\Delta H = mc_{p,avg} \Delta T$

This is a **reasonably accurate** way of determining Δu and Δh values for air and other ideal gases

- **NOTE: These equations are valid for ANY process**
- For a process from state 1 to state 2, evaluate the average values of the specific heats at the average temperature, T_{avg} .

$$T_{avg} = \frac{T_1 + T_2}{2}$$



Internal Energy, Enthalpy and Specific Heats for Ideal Gases

Specific Heat relations for Ideal Gases

- By combining the definition for enthalpy, $h = u + Pv$ and the ideal gas equation, $Pv = RT$ we get this important relation:

$$c_p = c_v + R$$

So for example, if we know c_p and R then we can determine c_v

- Also the specific heat ratio, k is defined as:

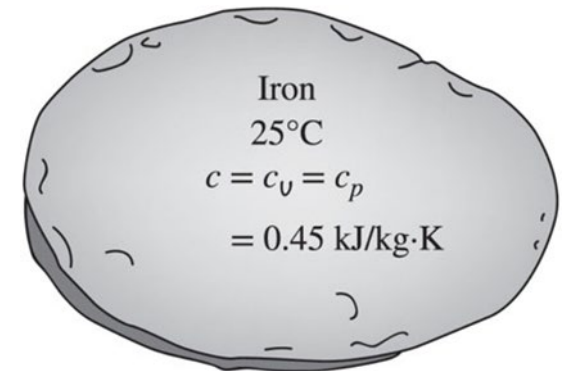
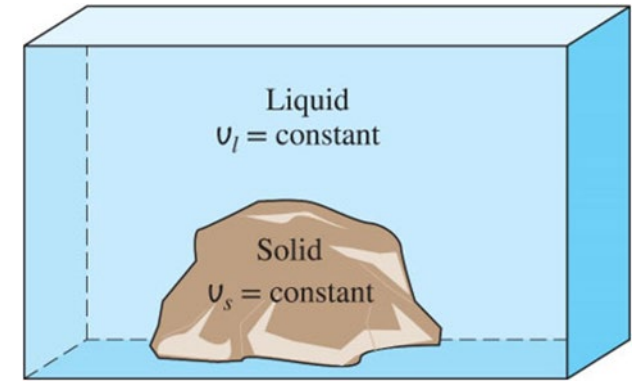
$$k = \frac{c_p}{c_v}$$

Internal Energy, Enthalpy and Specific Heats of Solids and Liquids

- A substance whose specific volume, v (or density) is constant with pressure is called an **incompressible substance**.
- The volumes of **solids and liquids** remain virtually unchanged during thermodynamic processes so they can be **approximated as incompressible substances** (with little loss in accuracy).
- It can be shown mathematically that c_v and c_p are identical for incompressible substances so we can drop the subscripts in these cases.

$$c_p = c_v = c$$

Note: in some tables of properties for solids and liquids the specific heat capacity is designated using c_p .



Internal Energy, Enthalpy and Specific Heats of Solids and Liquids

Internal energy changes

- The specific heat of incompressible substances is also a function of temperature alone therefore **for solids and liquids**:

$$\Delta u = c_{avg} \Delta T$$

$$\Delta U = mc_{avg} \Delta T$$

Enthalpy changes for solids and liquids

- For **solids**: $\Delta h = \Delta u \cong c_{avg} \Delta T$
- For **liquids** in a **constant pressure** process (e.g. heater): $\Delta h = \Delta u \cong c_{avg} \Delta T$
- For **liquids** in a **constant temperature** process (e.g. pump): $\Delta h = v \Delta P$

$$H = U + PV$$

where H is enthalpy, U is the internal energy, and P is pressure, and V is the volume.